# American Computer Science League

2025-2026 ● Contest 3: Short Problem Solutions ● Elementary Division

**1. Boolean Algebra**

x<100 AND x≥10, x≤99 AND x≥10, x<99 AND x>10**,** x≤99 AND x>9 represent the numbers from 10 to 99, but C doesn't include 10 or 99.

C. x<99 AND x>10

**2. Boolean Algebra**

$\sim(AB) * \sim(A + B)$ is TRUE when $\sim(AB)$ is TRUE and $\sim(A + B)$ is TRUE. $\sim(AB)$ is TRUE for all but (1,1). $\sim(A + B)$ is TRUE for just (0,0). Therefore, the overlap with the AND operator is (0,0).

A. (0,0)

**3. Boolean Algebra**

The Truth Table is given below so read top to bottom, it's 1101.

| $A$ | $B$ | $A + B$ | $\sim B$ | $\sim B * (A + B)$ | $\sim(\sim B * (A + B))$ |
|---|---|---|---|---|---|
| 0 | 0 | 0 | 1 | 0 | 1 |
| 0 | 1 | 1 | 0 | 0 | 1 |
| 1 | 0 | 1 | 1 | 1 | 0 |
| 1 | 1 | 1 | 0 | 0 | 1 |

B. 1101

**4. Boolean Algebra**

$\sim(\sim A + B) + \sim(A + \sim B) + \sim(A * B)$
$= A\sim B + \sim AB + \sim A + \sim B$
$= \sim B(A + 1) + \sim A(B + 1)$ using terms 1 and 4, then 2 and 3
$= \sim B(1) + \sim A(1)$
$= \sim B + \sim A$
$= \sim(B * A)$

A. $\sim(A * B)$

**5. Boolean Algebra**

$\sim(A + B)$ is (0,0). $\sim A * \sim B$ is just (0,0). $A + \sim B$ is (1,1), (1,0), (0,0). $\sim A + B$ is (1,1), (0,1), and (0,0) which is correct. $\sim A\sim B + AB$ is (0,0), (1,1). Only $\sim A + B$ is the same as $A \rightarrow B$.

D. $\sim A + B$

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|---|---|
| **6. Boolean Algebra**<br><br>If 20 students play both, then 30 students only play an instrument and 40 students only play a sport. Therefore, there are 10 other students that do neither. | E. 10 |